TG-WSPIS: Multi-Granular Text-Guided Weakly Supervised Pathology Image Segmentation

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***Abstract*—The high cost of pixel-level annotations has brought increasing attention to weakly supervised pathological image segmentation. However, conventional weak supervision approaches based on image-level labels often fail to capture the heterogeneity of cellular distributions, leading to incomplete supervision. To address this limitation, we propose TG-WSPIS, a text-guided weakly supervised pathological image segmentation paradigm that replaces traditional image-level labels with textual supervision. To bridge the visual-textual semantic gap, TG-WSPIS introduces a soft-alignment multimodal cell consensus algorithm and a unidirectional Gaussian mutual information-based collaborative learning mechanism, enabling more effective cross-modal alignment and improving segmentation performance. Specifically, we design three types of textual captions for each cell type in every image: global proportion, spatial distribution, and their fused representation, which are mapped into a unified multimodal semantic space. A newly proposed Pathology Multimodal Cell Consensus (PMCC) algorithm quantifies the semantic consistency between pathological images and textual descriptions. The framework further incorporates three caption-alignment branches and employs mutual learning to integrate cross-granularity knowledge. In addition, we construct two benchmark datasets, BCSS-T3G and WSSS4LUAD-T3G, to support comprehensive evaluation on the proposed TG-WSPIS task. Extensive experiments demonstrate that our method achieves superior segmentation performance across multiple metrics, confirming the effectiveness of text-guided weak supervision with multi-granular alignment. The dataset and source code will be released at: https://github.com/zms618/TG-WSPIS.**

***Index Terms*—Weakly Supervised Segmentation, Text-Guided Segmentation, Multimodal Learning, Computational Pathology**

I. Introduction

Accurate segmentation of diverse cell types in pathological images is critical for downstream tasks such as tumor diagnosis, pathological grading, and tumor microenvironment modeling [1]. For instance, the spatial distribution and relative proportions of distinct cell populations serve as critical indicators of tumor invasiveness and immune infiltration. While fully supervised methods have achieved remarkable success, they rely heavily on high-quality pixel-level annotations, the acquisition of which is prohibitively expensive (Fig. 1 a, b). Studies indicate that annotating a single full-resolution slide may require several hours of expert effort, while subjective factors further introduce substantial inter-expert variability, severely hindering large-scale clinical deployment [2]–[4]. Consequently, developing reliable weakly supervised segmentation methods has become a key focus in digital pathology.

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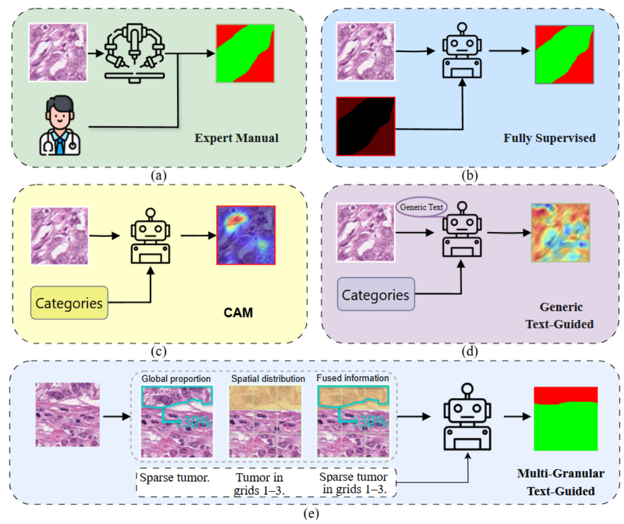

Fig. 1: Illustrations of four supervision paradigms for pathological image segmentation: (a) expert annotations, (b) full supervision, (c) CAM-based weak supervision, (d) generic text-guided weak supervision, and (e) our proposed multi-granular text guidance paradigm, which incorporates three levels of textual descriptions.

Weakly supervised pathological segmentation typically relies on class activation maps (CAM) to generate pseudo-labels [5]–[7]. However, CAM-based approaches tend to highlight only the most discriminative regions of a target class (Fig. 1 c), frequently neglecting broader contextual information and object completeness. To address this, recent approaches such as TPRO [8] incorporate text-guided supervision with generic class-level prompts (Fig. 1 d). However, such prompts are typically generic and fixed across images, and thus fail to capture the fine-grained, image-specific semantic variations among pathological samples.

To address these limitations, we propose TG-WSPIS, an image-level multi-granular text-guided paradigm for pathological image segmentation. Specifically, TG-WSPIS leverages textual supervision at three semantic granularities to guide more accurate and fine-grained pseudo-label generation through image-text alignment (Fig. 1 e). Unlike TPRO [8], which relies on generic textual cues describing cell types, our method constructs image-specific captions for each class. This design enables the model to capture individualized spatial distribution patterns and subtle semantic variations across pathological images.

However, this paradigm encounters two key challenges. First, there is a cross-modal semantic gap: pathological images contain dense and spatially dispersed information, whereas text descriptions are succinct and typically abstract. This substantial mismatch in information granularity makes establishing fine-grained semantic alignment between the two modalities particularly challenging [9]. Second, it is difficult to quantify spatial and numerical information: pathological analysis requires precise interpretation of continuous variations in cellular spatial distributions and relative abundances. These characteristics cannot be reduced to simple binary "presence/absence" indicators. Thus, both the magnitude of proportions and the precise locations need to be aligned. However, traditional contrastive learning or CLIP-based two-branch models focus primarily on global similarity and thus struggle to handle complex relationships involving both partial overlap and quantitative differences [10].

To overcome these challenges, inspired by advances in information theory and multimodal learning [11], we propose a weakly supervised segmentation framework with multi-granularity text guidance. Specifically, we establish three parallel alignment branches—global proportion, spatial distribution, and fused information—and devise a fine-grained vectorization scheme to embed both images and captions into a shared semantic space. Within this framework, we propose the Pathology Multimodal Cell Consensus (PMCC) algorithm to model per-cell soft consensus, enabling the capture of partial semantic overlaps. On this basis, we also incorporate a unidirectional Gaussian mutual information–based collaborative learning mechanism [12], using the fusion branch as the primary learner to absorb complementary semantics from the other branches.

Finally, we adopt the Grad-Eclip interpretability framework [13] to generate high-quality pseudo-labels from text-to-image attention maps. Extensive experiments show that TG-WSPIS achieves an mIoU of 76.79% on BCSS-T3G, exceeding the previous state-of-the-art ARML [14] by 5.23%. Even under minimal supervision using only class names, TG-WSPIS outperforms existing weakly supervised methods, validating the effectiveness of our multi-granular alignment framework.

Our main contributions are summarized as follows:

- We introduce a multi-granular text-guided weak-supervision paradigm for multi-class pathology image segmentation, where each image is paired with class-specific captions at three granularities (global proportion, spatial distribution, and fusion) to provide image-specific supervision without pixel-level annotations.
- We propose TG-WSPIS, a multi-granular text-guided weakly supervised segmentation framework that encodes multi-granular captions into structured semantic vectors, introduces PMCC for soft and quantifiable image-text alignment, and employs a unidirectional Gaussian mutual information collaborative learning strategy to fuse complementary cues across granularities for high-quality pseudo-mask generation.
- We build two standardized benchmarks, BCSS-T3G and WSSS4LUAD-T3G, by augmenting BCSS [15] and WSSS4LUAD [16] with three-level captions, and conduct extensive experiments on both benchmarks to validate the effectiveness of our method.

II. Benchmarks

To evaluate the proposed method, we construct two text-guided weakly supervised benchmarks, BCSS-T3G and WSSS4LUAD-T3G, derived from the public BCSS [15] and WSSS4LUAD [16] datasets, respectively. BCSS focuses on breast cancer with four tissue classes (tumor, stroma, lymphocyte, and necrosis), while WSSS4LUAD targets lung adenocarcinoma with three classes (tumor, stroma, and normal tissue).

Data Construction: Original whole-slide images (WSIs) are cropped into 224×224 patches, and dataset splitting is strictly performed at the WSI level to prevent data leakage. Guided by the CAP Cancer Protocol [17], we generate multi-granular textual supervision along two complementary dimensions. First, Anatomical Localizability, where each patch is partitioned into a 3×3 grid to describe spatial lesion distributions. Second, Diagnostic Quantifiability, where tissue area proportions are discretized into five ordered linguistic levels (rare, sparse, moderate, extensive, and diffuse) following semi-quantitative pathology conventions. Specifically, global area proportions are mapped to descriptive terms (e.g., 20%–40% $\rightarrow$ "sparse"), and interval midpoints are used as weak supervisory signals to model annotation uncertainty. All generated captions undergo expert review to ensure clinical validity.

The resulting benchmark statistics are summarized as follows: BCSS-T3G contains 28,000 patches (Train: 23,000; Val: 2,500; Test: 2,500), and WSSS4LUAD-T3G contains 28,000 patches with the same data split.

III. Method

This section presents the proposed TG-WSPIS framework for weakly supervised multi-class pathological cell segmentation. An overview of the architecture is shown in Fig. 2.

A. Fine-grained Vectorization and PMCC

Conventional contrastive learning adopts hard alignment, assuming zero similarity between non-matching image–text pairs. Such a binary assumption ignores the partial correlations that are ubiquitous in medical semantics, where pathological concepts often exhibit hierarchical relationships and semantic

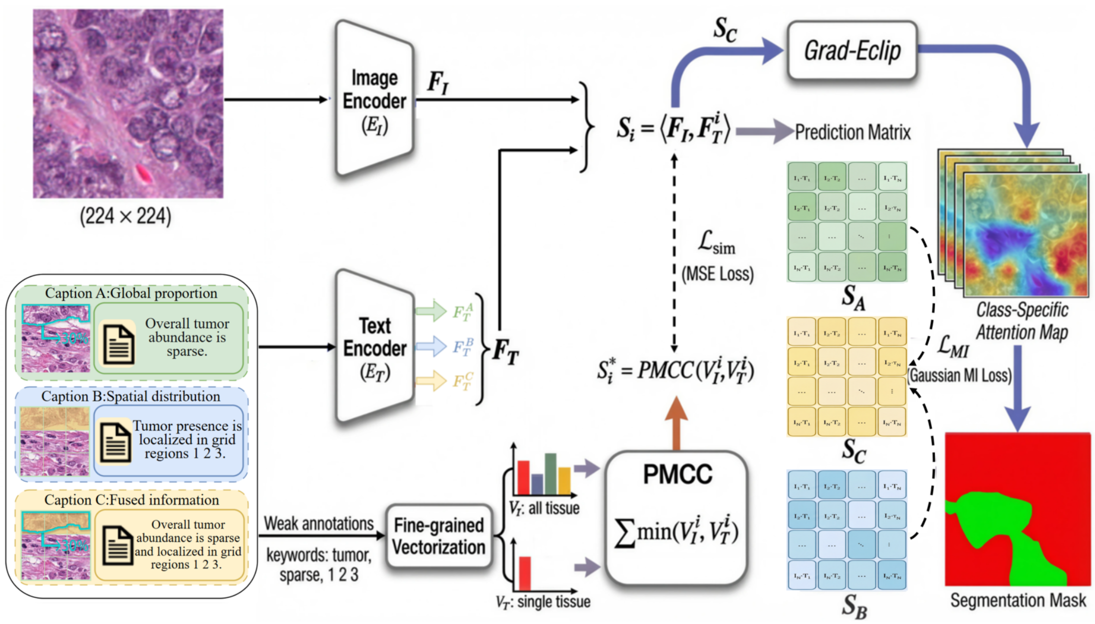

Fig. 2: Overall framework of the proposed TG-WSPIS.

overlaps. As a result, coarse alignment limits the quality of pseudo-label generation.

To address this limitation, we exploit a domain prior that pathological tissues consist of a finite set of cell populations. This property enables semantic information to be represented in a structured, fixed-length vector form. Accordingly, we design a unified vectorization scheme that maps image-level descriptions and per-cell-type textual descriptions into a shared d-dimensional space, preserving both spatial distribution and abundance information.

Formally, we define a unified semantic vector

$$\mathbf{V} = [\mathbf{v}^{(1)}; \mathbf{v}^{(2)}; \ldots; \mathbf{v}^{(K)}],$$

where $\mathbf{v}^{(k)}$ encodes the distribution of the k-th cell type. Each sub-vector corresponds to a 3×3 spatial grid, yielding a total dimensionality of $d = 9K$. Based on this scheme, we construct two types of vectors: Image Vector ($\mathbf{V}_I$): Generated by vectorizing the textual descriptions of all K cell types associated with an image and concatenating the resulting sub-vectors, representing multi-class semantic priors at the image level. Text Vector ($\mathbf{V}_T$): Generated by vectorizing a randomly sampled caption of a single cell type k, where only the corresponding sub-vector is activated and all others are zero-padded, serving as a class-conditional alignment target.

To quantify semantic consistency between $\mathbf{V}_I$ and $\mathbf{V}_T$, we propose the Pathology Multimodal Cell Consensus (PMCC) measure. Inspired by histogram intersection, PMCC computes the overlap between two distributions:

$$\mathrm{PMCC}(\mathbf{V}_I, \mathbf{V}_T) = \sum_{j=1}^{d} \min\left(\mathbf{V}_I[j], \mathbf{V}_T[j]\right), \qquad (1)$$

where $\mathbf{V}_I, \mathbf{V}_T \in \mathbb{R}^d$. Unlike hard alignment, PMCC explicitly captures partial agreement across cell types, spatial locations, and abundance levels.

To provide complementary alignment cues, we construct three granular textual descriptions for each cell type, each emphasizing a distinct semantic aspect:

Global Proportion: Encodes the overall abundance of a cell type (e.g., “moderate” $\rightarrow$ 50%), which is evenly distributed across all grid cells, yielding $\mathbf{v}_A = \frac{0.5}{9} \times [1, \ldots, 1]$.

Spatial Distribution: Encodes anatomical location within the 3×3 grid by activating the corresponding regions (e.g., grids 1–4), resulting in $\mathbf{v}_B = \frac{1}{9} \times [1, 1, 1, 1, 0, \ldots, 0]$.

Fused Information: Jointly encodes abundance and spatial distribution. Specifically, the qualitative abundance term (e.g., “moderate”) is first mapped to a global quantitative proportion (e.g., 50%). This proportion is then evenly distributed over the grid regions specified in the caption (e.g., grids 1–4), yielding $\mathbf{v}_C = \frac{0.5}{4} \times [1, 1, 1, 1, 0, \ldots, 0]$.

B. Multi-Granular Alignment Framework

To align pathological images with text at varying semantic levels, we develop a multi-branch framework based on CLIP [9]. This framework comprises three parallel branches, each dedicated to a specific caption granularity: global proportion (A), spatial distribution (B), and fused information (C).

We employ a pre-trained CLIP model as the backbone. Given an image I, the image encoder E_I extracts a visual feature vector $F_I = E_I(I)$. Similarly, the text encoder E_T processes the three textual descriptions for each cell type, yielding F_T^A, F_T^B, and F_T^C. The alignment target is derived using the PMCC algorithm. We compute the soft-alignment target similarity score S_i^* for each branch $i \in \{A, B, C\}$ as:

$$S_i^* = \mathrm{PMCC}(\mathbf{V}_I^i, \mathbf{V}_T^i) \qquad (2)$$

During training, the predicted cosine similarity $S_i = \langle F_I, F_T^i \rangle$ is supervised by S_i^* using the Mean Squared Error (MSE) loss:

$$\mathcal{L}_{\text{sim}}^i = \text{MSE}(S_i, S_i^*), \quad i \in \{A, B, C\} \tag{3}$$

The total alignment loss is the summation: $\mathcal{L}_{\text{sim}} = \sum_{i \in \{A,B,C\}} \mathcal{L}_{\text{sim}}^i$. This soft alignment objective enables the model to learn fuzzy matches and partial overlaps, which are critical for medical image analysis.

C. Gaussian Mutual Information Collaborative Learning

While the Global Proportion and Spatial Distribution branches effectively capture macroscopic semantic information, they lack precise boundary localization. Conversely, the Fused Information branch offers superior spatial discrimination but may suffer from limited semantic coverage due to the complexity of fused descriptions.

To synthesize the strengths of these branches, we introduce a Unidirectional Collaborative Learning mechanism driven by Gaussian Mutual Information (MI). This mechanism treats the Fused branch (C) as the primary learner, guiding it to absorb robust global semantics from branches A and B.

Let $S_A, S_B, S_C \in \mathbb{R}^{B \times B}$ denote the batch-wise similarity matrices predicted by the three branches. Assuming a Gaussian distribution for these predictions, we approximate the mutual information between the fused branch and the auxiliary branches to quantify their correlation:

$$\mathcal{L}_{\text{MI}}(S_i, S_C) = -\frac{1}{2} \ln\left(1 - \rho_{i,C}^2 + \epsilon\right), \quad i \in \{A, B\} \tag{4}$$

where ϵ is a stability constant, and $\rho_{i,C}$ is the Pearson correlation coefficient defined as:

$$\rho_{i,C} = \frac{\text{Cov}(S_i, S_C)}{\sqrt{\text{Var}(S_i)} \cdot \sqrt{\text{Var}(S_C)} + \epsilon} \tag{5}$$

This formulation encourages the Fused branch to maximize its statistical dependency with the global and spatial branches, thereby enforcing consistency across granularities. The total collaborative loss is:

$$\mathcal{L}_{\text{MI}} = \mathcal{L}_{\text{MI}}(S_A, S_C) + \mathcal{L}_{\text{MI}}(S_B, S_C) \tag{6}$$

The final objective function balances the alignment and collaborative losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sim}} - \eta\, \mathcal{L}_{\text{MI}} \tag{7}$$

where η is set to a small value to prevent the collaborative objective from dominating the optimization. This design is primarily driven by the image–text alignment objective, while the mutual information constraint acts as an auxiliary component to promote collaborative optimization among branches.

D. Pseudo-Label Generation

To generate high-quality pseudo-labels for segmentation, we utilize the fully trained Fused Information Branch, which integrates comprehensive semantic priors. The generation process follows three steps. First, compute the similarity scores between the input image and the K cell-type text descriptions using the Fused branch. Second, apply the Grad-Eclip framework to backpropagate gradients from the similarity scores to the image feature map, generating class-specific attention maps. Third, refine these attention maps into pixel-level pseudo-masks to supervise the downstream segmentation network.

IV. Experiments

A. Experimental Setup

Datasets and Metrics. We evaluate our method on the two benchmarks established in Sec. II: BCSS-T3G and WSSS4LUAD-T3G. Performance is assessed using four standard metrics: mean Intersection over Union (mIoU), Dice coefficient (mDice), Precision (mPrec), and Recall (mRec). All experiments are conducted under the same settings to ensure fair and consistent comparisons.

B. Comparison with State-of-the-Art Methods

We compare TG-WSPIS with recent state-of-the-art (SOTA) weakly supervised approaches. As shown in Tab. I, our method consistently achieves the best performance across both benchmarks.

TG-WSPIS achieves 76.79% and 75.35% mIoU on BCSS-T3G and WSSS4LUAD-T3G, respectively, achieving higher performance than the previous best method (ARML) [14] by 5.23% and 4.95%, respectively. These results demonstrate that our multi-granular textual alignment and collaborative learning strategy generalize well across different tissue types and organ domains.

C. Ablation Study and Analysis

To dissect the contribution of each component—specifically the PMCC soft-alignment and the Gaussian Mutual Information (MI) collaborative learning—we conduct a comprehensive ablation study on the BCSS-T3G dataset, as detailed in Tab. II. Effectiveness of PMCC Soft Alignment. Standard contrastive learning (Hard Alignment, Rows 1-4) yields suboptimal results (mIoU $\approx$ 61-64%), as it enforces binary matching that contradicts the continuous nature of pathological semantics. In contrast, substituting Hard Alignment with our proposed PMCC (Rows 5-9) brings a dramatic performance leap, boosting mIoU by over 10%. This confirms that PMCC's ability to model partial semantic overlaps is fundamental to the framework's success.

Robustness under Minimal Supervision. A critical observation lies in the comparison with existing WSSS methods under the same supervision budget. The configuration "PMCC + Class Info" (Row 5) utilizes only category names as textual input, keeping the supervision strength consistent with other baseline methods. Remarkably, even without detailed descriptions, this baseline achieves a mIoU of 72.79%, already outperforming the SOTA method ARML (71.56%) by 1.23%.

This proves that our performance gain stems primarily from the superior alignment architecture rather than solely from richer textual annotations.

TABLE I: Quantitative comparison with state-of-the-art methods on the BCSS-T3G and WSSS4LUAD-T3G datasets.

	BCSS-T3G								WSSS4LUAD-T3G						
Method	**TUM**	**STR**	**LYM**	**NEC**	**mIoU**	**mDice**	**mPrec**	**mRec**	**TUM**	**STR**	**NOM**	**mIoU**	**mDice**	**mPrec**	**mRec**
HistoSegNet [18]	34.44	47.33	30.14	28.83	35.19	51.64	52.12	51.17	32.79	45.68	31.66	36.71	53.40	54.05	52.76
SEAM [19]	74.53	60.92	56.24	61.82	63.38	77.38	78.02	76.75	67.22	57.26	63.38	62.62	76.94	77.58	76.31
WSSS-Tissue [20]	72.94	67.34	56.22	59.24	63.94	77.80	78.45	77.16	73.58	60.12	54.45	62.72	76.79	77.42	76.17
OEEM [21]	76.11	69.88	51.34	60.07	64.35	77.90	78.58	77.23	72.14	60.73	56.84	63.24	77.29	77.95	76.64
HAMIL [22]	73.68	69.64	57.51	57.71	64.64	78.29	78.85	77.74	76.59	61.36	52.24	63.40	77.14	77.82	76.47
SAM3 [23]	76.65	67.98	61.05	53.65	64.83	78.34	79.05	77.64	66.76	62.23	60.58	63.19	77.41	78.08	76.75
TPRO [8]	74.61	66.82	60.50	70.23	68.04	80.87	81.35	80.40	71.10	66.30	65.80	67.73	80.74	81.44	80.05
ARML [14]	79.10	73.67	64.62	68.83	71.56	83.31	83.92	82.71	74.56	70.62	66.03	70.40	82.58	83.18	81.99
Ours	**84.33**	**77.44**	**70.68**	**74.71**	**76.79**	**86.78**	**87.54**	**86.04**	**79.43**	**75.08**	**71.55**	**75.35**	**85.91**	**86.68**	**85.15**

Impact of Multi-Granular Guidance & MI. While single-granularity captions (Global/Spatial) achieve competitive results (Rows 6-7), the Fused Information (Row 8) further integrates these cues, reaching 75.65%. Finally, incorporating the Gaussian MI Collaborative Learning (Row 9) yields the best performance (76.79%). By forcing the fused branch to absorb global and spatial consistency from auxiliary branches, MI effectively bridges the semantic gap, providing an additional gain of 1.14% over the standalone fused branch.

TABLE II: Ablation study on the BCSS-T3G dataset. Hard: Standard Hard Alignment; PMCC: Our Soft Alignment; MI: Gaussian Mutual Information Collaborative Learning.

Method	**mIoU**	**mDice**	**mPrec**	**mRec**
Hard + Global Proportion	62.23	76.71	77.45	75.99
Hard + Spatial Distribution	61.78	76.38	77.12	75.66
Hard + Fused Information	62.55	76.96	77.70	76.24
Hard + Fused Info + MI	63.54	77.71	78.45	76.99
PMCC + Class Name Only	72.79	84.25	85.10	83.42
PMCC + Global Proportion	74.89	85.64	86.45	84.85
PMCC + Spatial Distribution	74.49	85.38	86.20	84.58
PMCC + Fused Information	75.65	86.14	86.95	85.35
PMCC + Fused Info + MI (Ours)	**76.79**	**86.79**	**87.54**	**86.04**

D. Hyperparameter Sensitivity on η

We investigate the impact of the weighting coefficient η in Eq. (7), which balances the alignment and mutual information losses. Experiments are conducted on the BCSS-T3G dataset under the same settings as Sec. IV-A. As shown in Tab. III, TG-WSPIS achieves the highest mIoU of 76.79% at $\eta = 0.0001$. When η is too large, the collaborative term dominates and disrupts alignment consistency, leading to suboptimal performance; conversely, overly small values hinder effective knowledge transfer. This demonstrates that TG-WSPIS remains stable within a small range of η and achieves optimal performance with a balanced loss contribution.

TABLE III: Sensitivity of TG-WSPIS to the weighting coefficient η on BCSS-T3G.

η	0	0.00001	**0.0001**	0.001	0.01
mIoU (%)	75.65	76.01	**76.79**	75.56	69.21

E. Analysis of Image–Text Alignment

To visually and quantitatively validate why our method works, we analyze the alignment accuracy. A core hypothesis of our framework is that the proposed PMCC mechanism yields a softer, more precise semantic alignment than standard contrastive learning. To verify this, we compute the Mean Squared Error (MSE) between the predicted image–text consensus scores and the ideal alignment targets on the validation set.

As detailed in Tab. IV, the conventional Hard Alignment strategy (employed in standard CLIP) results in high MSE values (> 0.26), indicating a struggle to capture the continuous nuances of pathological semantics. In stark contrast, our PMCC-based soft alignment achieves negligible error rates (e.g., 0.005 for Global Proportion), reducing the MSE by an order of magnitude. This quantitative evidence confirms that PMCC effectively bridges the semantic gap, enabling the model to learn fine-grained consensus between histological images and multi-granular textual descriptions.

TABLE IV: Quantitative analysis of alignment consistency.

Feature Granularity	**Alignment Strategy**	**MSE (↓)**
Global Proportion	Hard Alignment	0.269
Spatial Distribution	Hard Alignment	0.407
Fused Information	Hard Alignment	0.275
Collaborative Learning	Hard Alignment	0.331
Global Proportion	**PMCC (Ours)**	**0.005**
Spatial Distribution	**PMCC (Ours)**	**0.013**
Fused Information	**PMCC (Ours)**	**0.048**
Collaborative Learning	**PMCC (Ours)**	**0.040**

F. Evaluation of Pseudo-Label Quality

The ultimate goal of WSSS is to generate high-quality pseudo-masks that can serve as reliable ground truth for downstream fully supervised models. To assess this utility, we conducted a "proxy task" experiment: we generated pseudo-labels using different WSSS methods on the training set and then trained a standard U-Net from scratch using these pseudo-labels. The models were evaluated on the BCSS-T3G test set using expert annotations.

Table V reports the segmentation performance. The U-Net trained with our pseudo-labels achieves a remarkable mIoU of 77.26%. This performance significantly surpasses the models trained with pseudo-labels from SOTA methods, outperforming ARML by 5.43% and TPRO by 9.11%. These results strongly indicate that our framework produces dense, pixel-precise supervision signals.

TABLE V: Evaluation of pseudo-label quality.

Source of Pseudo-Labels	TUM	STR	LYM	NEC	mIoU
TPRO [8]	74.65	66.90	60.65	70.40	68.15
ARML [14]	79.30	73.95	64.90	69.15	71.83
Ours	**84.52**	**77.98**	**71.44**	**75.10**	**77.26**

G. Qualitative Analysis

As shown in Fig. 3, TG-WSPIS generates coherent masks with sharp boundaries, effectively segmenting scattered tissue regions missed by prior methods.

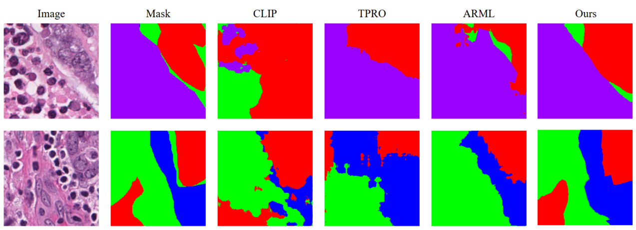

Fig. 3: Qualitative comparison of multi-class cell segmentation results on the BCSS-T3G dataset.

V. Conclusion and Future Work

We present TG-WSPIS, a weakly supervised pathological image segmentation framework driven by multi-granular text guidance. It aligns images with global proportion, spatial distribution, and fused textual information via the proposed PMCC algorithm, while a GMI-based collaborative learning strategy enhances cross-granular consistency.

Experiments on BCSS-T3G and WSSS4LUAD-T3G show that TG-WSPIS outperforms state-of-the-art methods and remains robust under minimal supervision.

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